

Engine Only Operates in Idle Mode

Symptoms

- The engine will **only** operate in idle mode.

How To Use This Tree

This symptom tree can be used to troubleshoot engine idle symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check Idle/Rated Switch Signal Wire for Short Circuit	Less than 10 ohms resistance?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	
	STEP 2A. Check Idle/Rated Switch Signal Wire	Less than 10 ohms resistance?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check Idle/Rated Switch Signal Wire for Short Circuit

Conditions : • Open the customer interface box • Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.		
Action	Specification/Repair	Next Step

Check idle/rated switch signal wire. • Disconnect the idle/rated switch signal wire from the X4 connector. • Place one test lead on the idle/rated switch signal pin in connector C3. • Place the other test lead on the battery 1 voltage return terminal of the customer interface box logic unit.	Less than 10 ohms resistance? YES Repair : Replace the wire(s). Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	Repair complete.
	Less than 10 ohms resistance? NO	2A

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check Idle/Rated Switch Signal Wire

Conditions :

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C10 from the engine harness.

Action	Specification/Repair	Next Step

Check idle/rated switch signal wire. • Place one test lead in the idle/rated switch signal pin of the C3 connector. • Place the other test lead on another pin in the C3 connector. • Repeat for all other pins in the C3 connector.	Less than 10 ohms resistance? YES Repair : Refer to Section TF in the Troubleshooting and Repair Manual, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493, or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60, CM850 Modular Common Rail System Series Engines, Bulletin 4021533, or refer to the OEM service manual for idle switch repair instructions.	Repair complete.
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.